

Question: 1

Convert the following IP address to its 32-bit binary representation: 172.16.5.10. Write out each octet in binary, separated by dots.

Answer: 1

Binary Conversion Table (172.16.5.10)

To convert the IP address to its **32-bit binary representation**, each decimal number (octet) is converted into an 8-bit binary block:

Decimal Octet	Binary Conversion Calculation	8-Bit Binary Representation
172	$128 + 32 + 8 + 4$	10101100
16	16	00010000
5	$4 + 1$	00000101
10	$8 + 2$	00001010

Final 32-Bit Binary Representation

10101100.00010000.00000101.00001010

Question: 2

Given the binary IP address 11000000.10101000.00000001.00001111, convert it back to standard dotted decimal notation.

Answer: 2

Binary to Decimal Conversion Table

To convert the 32-bit binary IP address back to standard **dotted-decimal notation**, each 8-bit binary block (octet) is converted by adding the positional values of the 1 bits:

Binary Octet	Positional Math Conversion	Decimal Value
11000000	$128 + 64$	192
10101000	$128 + 32 + 8$	168
00000001	1	1
00001111	$8 + 4 + 2 + 1$	15

Final Dotted-Decimal Notation

192.168.1.15

Question: 3

You are given a list of 10 IP addresses used by different apps (e.g., 192.168.43.1, 8.8.8.8, 10.0.5.25, 172.20.14.2, 203.0.113.5, 172.15.255.255, 192.0.2.10, 192.168.1.100, 123.45.67.89, 172.16.0.1). For each, identify if it is a private or public IP address and explain your reasoning.

Answer: 3

Private vs Public IP Address Classification Table

To classify each IP address, we compare it against the standard RFC 1918 private address ranges:

- **Class A Private Range:** 10.0.0.0 to 10.255.255.255
- **Class B Private Range:** 172.16.0.0 to 172.31.255.255
- **Class C Private Range:** 192.168.0.0 to 192.168.255.255

#	IP Address	Classification	Reasoning / Explanation
1	192.168.43.1	Private	Falls perfectly inside the Class C private block (192.168.0.0/16) commonly used for local router hot-spots.
2	8.8.8.8	Public	Sits completely outside all private ranges. It is a globally routable public address (Google's Public DNS server).
3	10.0.5.25	Private	Fits right inside the standard Class A private IP space (10.0.0.0/8) used for large internal networks.
4	172.20.14.2	Private	Falls within the Class B private range, as the second octet (20) is between the private boundaries of 16 and 31.
5	203.0.113.5	Public	Outside all private address blocks. It belongs to a globally unique internet registry block.
6	172.15.255.255	Public	Public IP. Even though it starts with 172, the second octet (15) falls below the private Class B threshold which starts at 16.
7	192.0.2.10	Public	Outside the private ranges. The Class C private block only covers addresses starting with 192.168.
8	192.168.1.100	Private	Fits directly into the Class C private space (192.168.0.0/16) routinely assigned to home devices.
9	123.45.67.89	Public	Sits entirely outside all RFC 1918 private address specifications and can be routed anywhere across the internet.

10	172.16.0.1	Private	Fits exactly at the very beginning boundary of the Class B private allocation pool (172.16.0.0/12).
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Question: 4

For the CIDR notations /20, /25, and /30, calculate the total number of usable IP addresses in each subnet. Show your working for each calculation.
 Usable addresses = $(2^{\text{number of host bits}}) - 2$ (for network and broadcast addresses).

Answer: 4

Subnet Usable IP Calculations Table

An **IPv4 address has exactly 32 bits** in total. To find the total number of usable host IP addresses, we calculate the remaining host bits using the formula:

$$\text{Host Bits} = 32 - \text{CIDR Subnet Prefix}$$

$$\text{Total Usable Addresses} = (2^{\text{Host Bits}}) - 2$$

(We subtract 2 because the very first address is the Network ID and the very last address is the Broadcast Address).

CIDR Prefix	Host Bits Calculation	Total IP Pool ($2^{\text{Host Bits}}$)	Subtract 2 Rule	Total Usable IP Addresses
/20	$32 - 20 = 12 \text{ bits}$	$2^{12} = 4,096$	$4,096 - 2$	4,094 usable IPs
/25	$32 - 25 = 7 \text{ bits}$	$2^7 = 128$	$128 - 2$	126 usable IPs
/30	$32 - 30 = 2 \text{ bits}$	$2^2 = 4$	$4 - 2$	2 usable IPs

Detailed Step-by-Step Working

- 1. For /20 Subnet:
 - Host Bits:** $32 - 20 = 12$ bits left for local network assignments.
 - Total Addresses:** $2^{12} = 4,094$ total absolute mathematical addresses.

- **Usable IPs:** $4,096 - 2 = 4,094$ devices can safely register on this block.
- **2. For /25 Subnet:**
 - **Host Bits:** $32 - 25 = 7$ bits left for local network assignments.
 - **Total Addresses:** $2^7 = 128$ total absolute mathematical addresses.
 - **Usable IPs:** $128 - 2 = 126$ devices can safely register on this block.
- **3. For /30 Subnet:**
 - **Host Bits:** $32 - 30 = 2$ bits left for local network assignments.
 - **Total Addresses:** $2^2 = 4$ total absolute mathematical addresses.
 - **Usable IPs:** $4 - 2 = 2$ devices can safely register on this block (*commonly used for direct point-to-point router links*).

Question: 5

Imagine you are setting up a small LAN for a gaming night with friends using your WiFi router at home. Assign static private IP addresses to 5 devices (laptop, phone, tablet, smart TV, and PlayStation) using the 192.168.0.0/24 range. List the IP you would assign to each device and explain why you chose that range.

Answer: 5

Static IP Address Allocation Table (192.168.0.0/24)

For a local gaming night network, the home Wi-Fi router acts as the default gateway (usually assigned the very first usable IP address: 192.168.0.1). The 5 devices are assigned **static private IP addresses** outside of the router’s automatic DHCP pool to prevent address conflicts:

Device Type	Assigned Static IP Address	Subnet Mask	Default Gateway	Role / Placement Choice
Wi-Fi Router	192.168.0.1	255.255.255.0	—	Core Gateway device routing all internal network traffic.
PlayStation	192.168.0.10	255.255.255.0	192.168.0.1	Primary gaming console; given a low, easy-to-remember IP.
Gaming Laptop	192.168.0.11	255.255.255.0	192.168.0.1	Main multiplayer host client machine.

Smart TV	192.168.0.20	255.255.255.0	192.168.0.1	Media screen display client; grouped in the .20 series.
Smartphone	192.168.0.30	255.255.255.0	192.168.0.1	Companion chat device; grouped out of the gaming pool.
Tablet	192.168.0.31	255.255.255.0	192.168.0.1	Dashboard/strategy client machine.

Justification for Choosing the 192.168.0.0/24 Range

- **Standard Class C Private Address Space:** The 192.168.0.0 network block is defined by RFC 1918 strictly for local area networking (LAN). It cannot be routed on the public internet, making it entirely secure and free to use internally without registering it with an internet authority.
 - **Perfect Scale for Home Use:** The /24 subnet prefix (Subnet Mask: 255.255.255.0) provides exactly **254 usable IP addresses** (192.168.0.1 to 192.168.0.254). This pool easily fits a gaming night setup, leaving plenty of free numbers for friends to join without wasting a massive block of unassigned network spaces.
 - **Native Router Compatibility:** Almost all commercial home routers ship with 192.168.0.0 or 192.168.1.0 pre-configured as their default internal network layout. Sticking to this range eliminates the need to change complex underlying router system configurations, making setup quick and hassle-free.
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